Signal Portfolio Construction via Modified Determinant Model

MScFE Capstone Project · WorldQuant University · June 2026

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PRESENTERS

Group Information

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Project Goal & Motivation

Two Core Problems We Address

1. Non-Synchronous Trading Bias

Daily correlations understate true co-movement for globally-spread indices (e.g. S&P 500 vs. Nikkei 225) due to non-overlapping trading hours.

2. Regime Instability & Diversification Breakdown

In March 2020 most global portfolios fell about 33% and cross-country diversification failed precisely when needed most.

Our Goal

Develop a correlation-based regime detection signal (KIKO) to improve portfolio construction across 7 global equity indices

7 Global Equity Indices — 2015 to 2025

S&P 500 (USA)
B3-Bovespa (Brazil)
FTSE 100 (UK)
EURO STOXX 50 (Europe)
FTSE/JSE Top 40 (S. Africa)
Nikkei 225 (Japan)
ASX 50 (Australia)

LITERATURE REVIEW

Key Research Foundations & Gaps

Noise in Correlation Matrices

Laloux et al. (1999) & Plerou et al. (1999): most eigenvalues in financial correlation matrices are statistical artifacts (RMT). Optimizers exploit spurious correlations, failing out-of-sample.

Multi-Scale & Non-Synchronous Trading

Fernandez-Macho (2012): European markets appear far more integrated at weekly than daily horizons — a non-synchronous artifact, not genuine independence.

Dynamic Correlations & Crisis Contagion

Engle (2002) DCC; Preis et al. (2012): during downturns, pairwise correlations narrow and shift upward — diversification benefits disappear precisely when most needed.

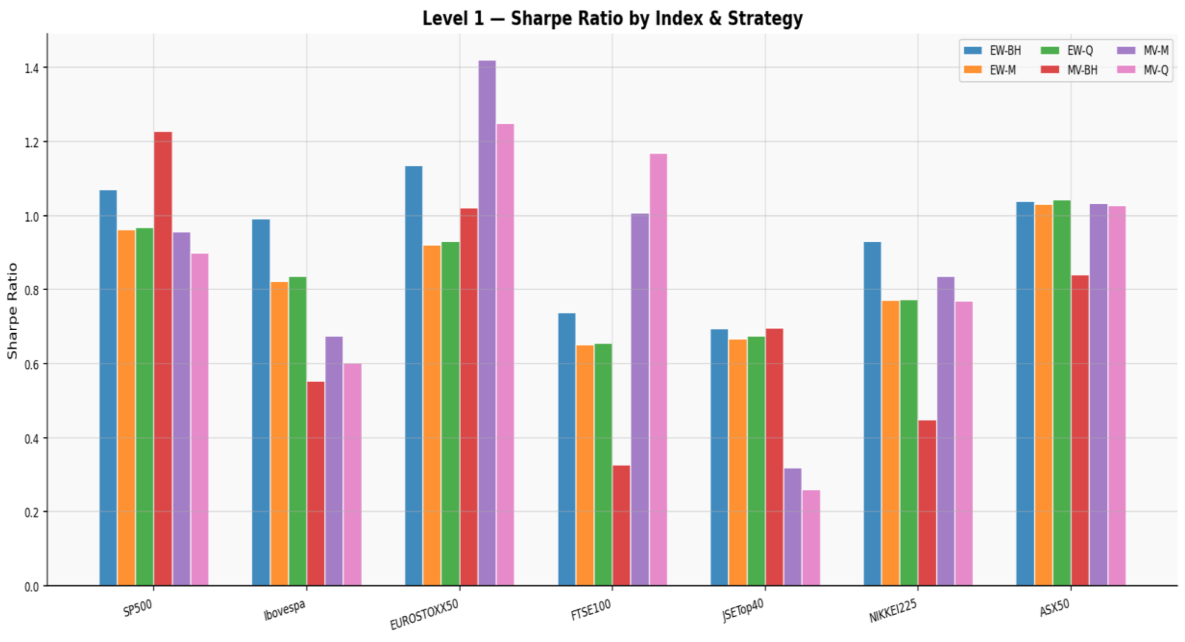
Network & Hierarchical Approaches

Mantegna (1999) MST; Lopez de Prado HRP: hierarchical clustering for portfolio weights avoids covariance matrix inversion, reducing estimation error in large universes.

No systematic evidence on how correlations change across horizons for a geographically diverse, multi-timezone index set — or how that structure can drive buy/sell signals.

BENCHMARK PORTFOLIOS

Sharpe Ratios & Cumulative Returns



Level-2 Global Portfolios

G1 EW

G2 MV

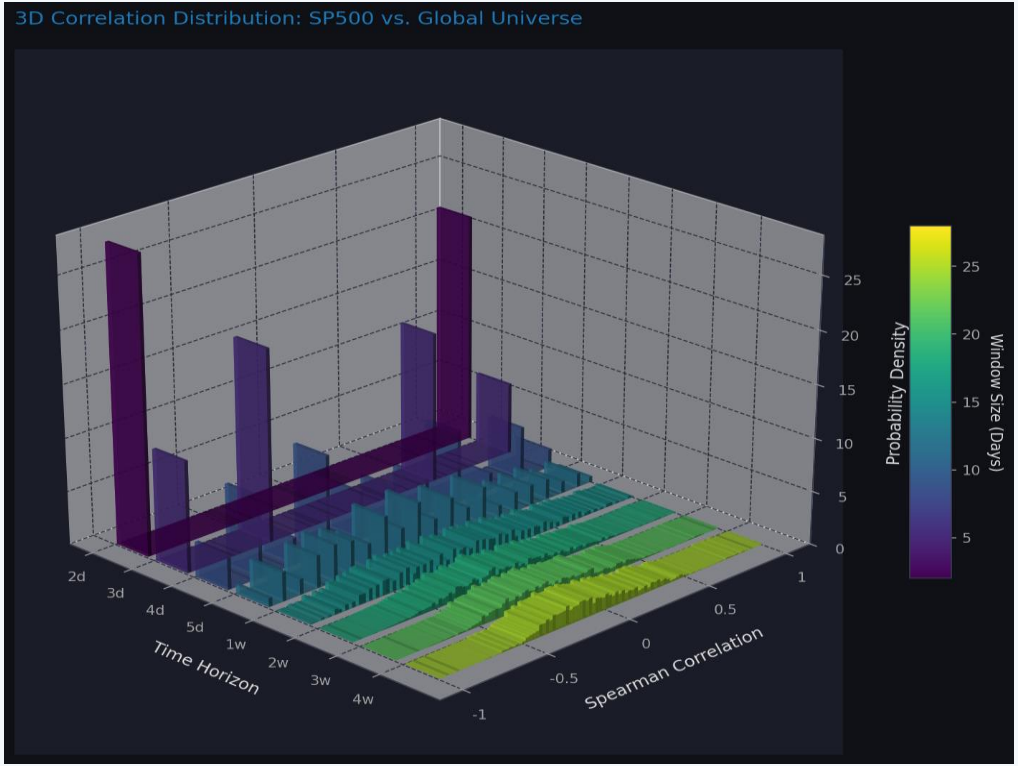
G3 RP

G4 HRP

Key Finding: EW rebalancing frequency is irrelevant for large N — EW-BH ≈ EW-M ≈ EW-Q

EXPLORATORY DATA ANALYSIS

Return Distributions & Rolling Correlations



Key Finding: Correlations are highly oscillatory and non-stationary – motivating for multi-scale analysis (3d, 7d, 28d windows).

MAIN CONTRIBUTION

KIKO — The Modified Determinant Model

Theoretical Motivation

Let $R(w)$ be the rolling correlation matrix. We compute a modified determinant, zeroing the diagonal to isolate off-diagonal structures:

Key idea: $\det(R - I) = \prod(\lambda_i - 1)$

- if $\det \to 0$: eigenvalue ≈ 1
  $\to$ stress, coupled regime
- Large $|\det|$: eigenvalues far from unity
  $\to$ calm, diversified

This scalar captures systemic co-movement and transitions between coupled (risk-on) and decoupled (risk-off) regimes.

Signal Construction Pipeline

1. **Rolling Correlation Matrices**

   Compute $R(w)$ for $w \in \{3,7,28\}$ days from constituent stock log-returns for each index.

2. **Modified Determinant**

   Calculate $\det(R - I)$: removes trivial diagonal, exposes pure co-movement structure.

3. **Z-Score Standardization**

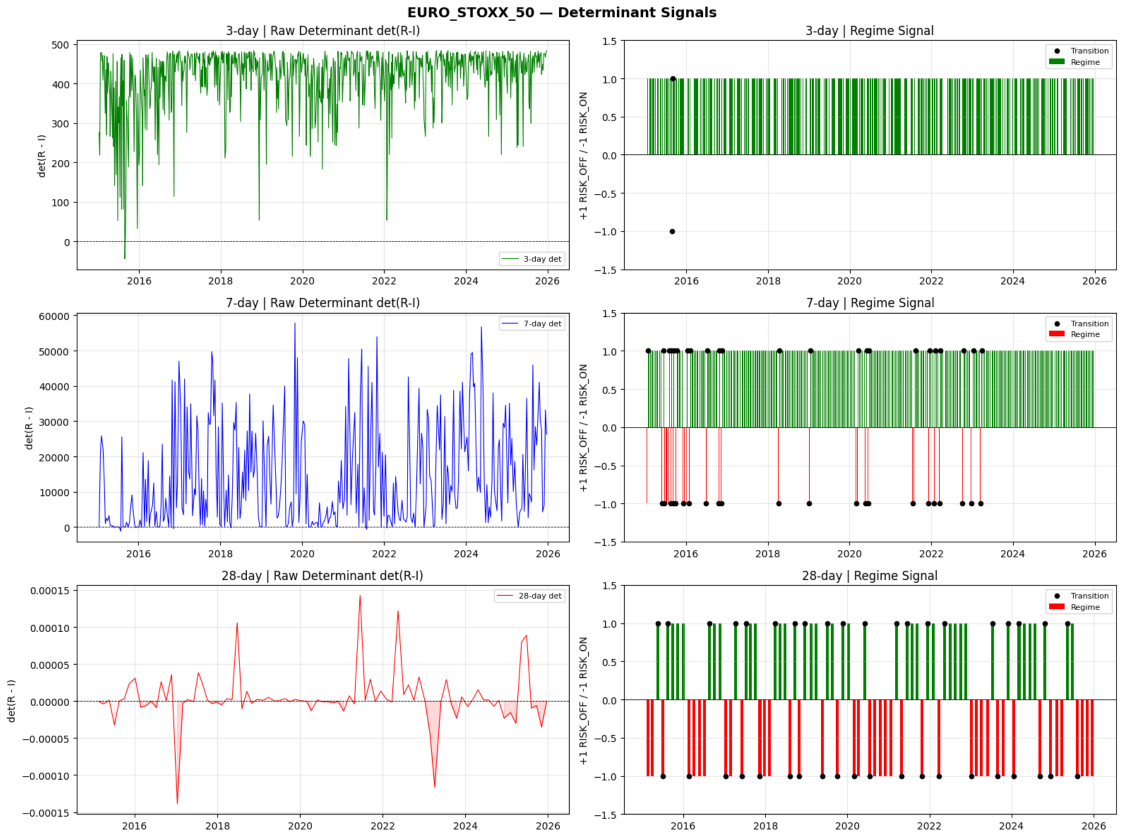
   Standardize using 252-day rolling window: $z = (\det - \mu) / \sigma$

4. **Composite Signal**

   Average z-scores across 3 windows $\to$ $z\_composite = (z_3 + z_7 + z_{28}) / 3 \to$ Buy/Sell.

KIKO RESULTS

Modified Determinant Signal — Raw Values & Regime Transitions



Modified Determinant: det(R–I) Raw Values (log scale) across different windows:

3-day (green)

7-day (blue)

28-day (red)

Z-Score Signals:

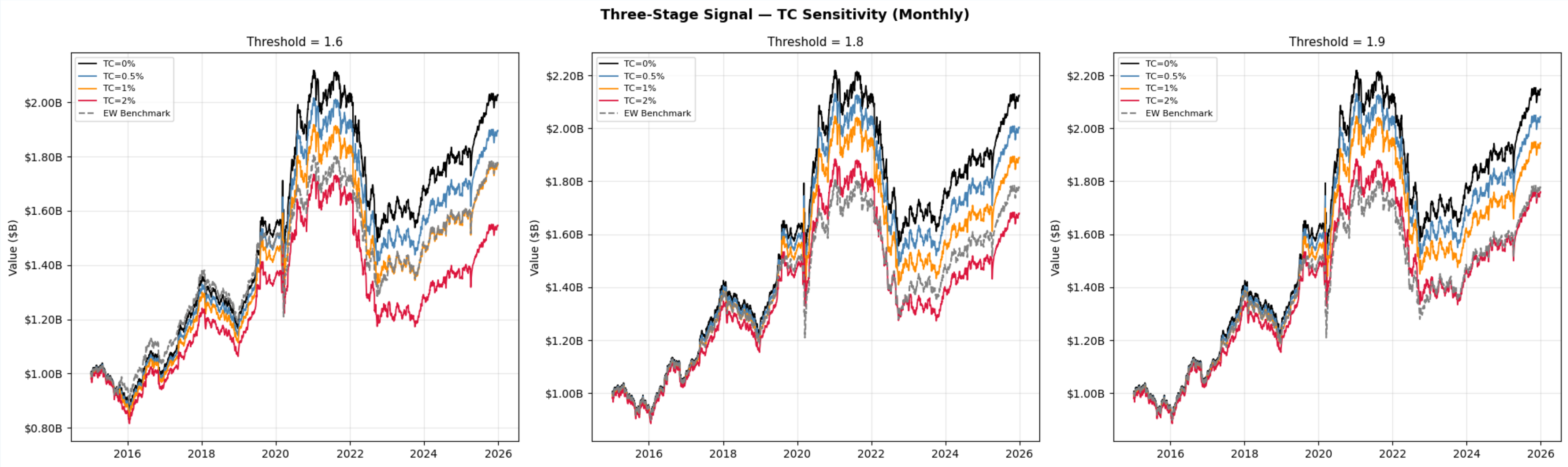
+1 = Decoupled (Risk-OFF)

−1 = Coupled (Risk-ON).

Red = transitions to negative

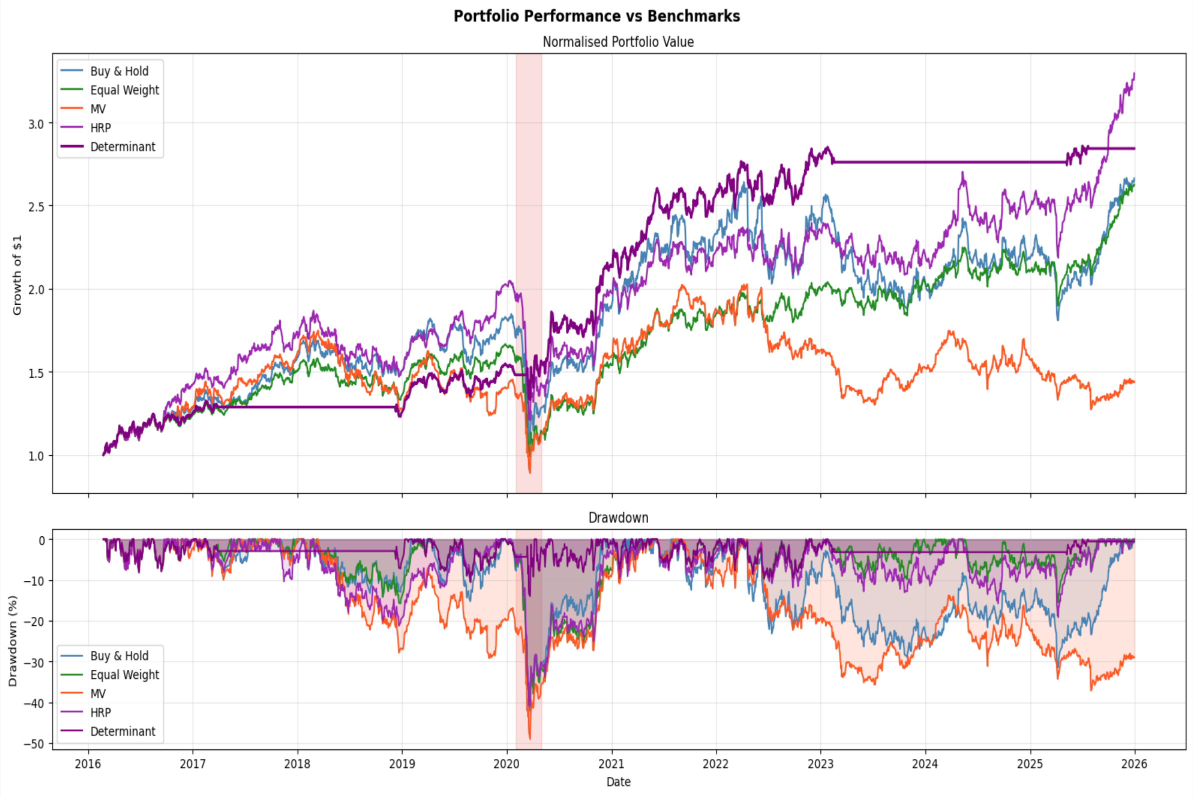
Green = transitions to positive.

Portfolio Performance based on KIKO including transactions costs vs. EW



RESULTS

Portfolio Performance of Binary Signal regime vs. Benchmarks



Strategy	Sharpe	MaxDD	CAGR
Determinant	**0.976**	**-13.96%**	**12.96%**
HRP	0.783	-40.91%	14.92%
Equal Weight	0.729	-40.47%	11.92%
Buy & Hold	0.620	-42.12%	12.09%
MV	0.313	-49.07%	4.34%

KIKO - Key Advantages:

- Smallest Drawdown compared to benchmarks
- Binary regime approach correctly exited market before COVID-19 crash (pink band, 2020) and re-entered after recovery.

Summary & Research Outlook

Key Findings

- Mean pairwise correlation rises:
  0.252 → 0.547 (daily → 4-week)
  non-synchronous bias is large and measurable.
- KIKO binary regime:
  Sharpe 0.976 vs. 0.620 (Buy & Hold)
  with Max DD of only −13.96% vs. −42%.
- Inverted-signal test confirms KIKO captures genuine structural market information.
- COVID-19 sell-off correctly identified
  capital protected — key real-world validation.

Limitations

- Full-sample strategy selection introduces look-ahead bias
- USD-denominated only - FX exposure not hedged
- Proportional TC model ignores market impact

Future Directions

- Run an exhaustive rolling-window parameter search all possible combinations: 1 day – 30 days
- Add hedging, short-selling and leverage to the KIKO approach from a risk management viewpoint
- ML/deep-learning extensions using KIKO as input feature

PEER REVIEW

Critical Feedback & How We Addressed It

Feedback: Objective statement too broad

Reviewer noted the objective statement needed more precision and should serve as the main guide.

Our Response

We clarified the primary goal in the abstract: detect regime transitions via det(R−I) and improve Sharpe vs. benchmarks.

Feedback: Research gap not clearly identified

The current research related to this topic was not explained; literature review failed to identify research gaps.

Our Response

We expanded Section 2 to explicitly note that no systematic multi-horizon, multi-timezone study existed.

Feedback: Scope too broad, unclear model selection

Methodology well-written but exaggerated; unclear model selection criteria and validation strategy.

Our Response

We added an extensive parameter scan (heatmaps, Fig. 18 & 19) and binary regime validation as a focused, falsifiable hypothesis.

Feedback: No conclusion in proposal

The proposal did not include a conclusion; only a proposed outcome for portfolio performance.

Our Response

Final report has a full Conclusion & Outlook section (Sec. 7) with quantitative results, limitations, and future research.

PERSONAL CONTRIBUTIONS

Who Did What

KB Kristian Boroz

- KIKO operator design & theoretical motivation
- Modified determinant signal construction pipeline
- Parameter optimization (1,000+ combinations)
- Binary regime classification framework
- GitHub repository structure & code integration

IKS Isaac Kiplimo Sum

- ISAAC framework: adaptive correlations regime detection
- EDA: stationarity tests, rolling correlations, copula analysis
- Multi-scale horizon selection methodology
- Level-1 benchmark portfolio simulation engine
- Performance metrics computation & reporting

ITB Israel Twum Bosompem

- I-WAN framework: global Level-2 portfolio overlays
- G1–G4 global strategy construction & analysis
- Volatility-targeting overlay (G4-VT) implementation
- G5 macro-enhanced signal design
- Literature review: network & HRP approaches

Thank You

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GitHub: github.com/kboroz/MSFE_Capstone_Project

We thank Professor Byers and WorldQuant University for the guidance, brilliant program, and the foundation for a new stage in our professional careers.